

Notes: Koudijs & Voth (AER, 2016)

Leverage and Beliefs: Personal Experience and Risk-Taking in Margin Lending

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1 Big picture

- **Question.** What determines risk-bearing capacity and leverage in financial markets? The paper asks whether lenders' personal experience with a financial scare can change beliefs and risk-taking, even when lenders do not actually lose money.
- **Setting.** The paper studies eighteenth-century Amsterdam margin lending. Loans were secured by traded securities, especially East India Company (EIC) stock, and loan contracts are observed in notarial archives.
- **Natural event.** Around Christmas 1772, the Seppenwolde syndicate collapsed after speculating in EIC shares. Some lenders had financed the syndicate and were forced to liquidate collateral. They were at risk of losses but ultimately recovered principal and interest.
- **Treatment variation.** Exposed lenders had direct personal exposure to the syndicate. Non-exposed lenders saw the same public market turmoil but did not personally face potential losses from the syndicate's loans.
- **Main result.** After Christmas 1772, exposed lenders demanded substantially higher haircuts than non-exposed lenders. The benchmark estimates imply a rise of roughly 4–8 percentage points in haircuts, a large increase relative to a pre-crisis haircut around 20 percent.
- **Interest-rate result.** Interest rates do not increase in the same way. The adjustment is primarily through collateral requirements rather than loan rates.
- **Behavioral channel.** The paper interprets the result as a change in beliefs or risk-taking caused by personal experience. Exposed lenders behave more pessimistically even though the event was public and they did not suffer realized losses.
- **Aggregate relevance.** Major exposed lenders were important players in the Amsterdam collateralized lending market. Their changed behavior explains a meaningful market-wide decline in leverage.
- **Robustness.** The result survives lender/borrower fixed effects, borrower-time fixed effects, controls for borrower riskiness, relationship-capital tests, matching estimators, lender-type interactions, and attrition checks.
- **Takeaway.** Leverage cycles can arise from changes in lenders' beliefs and risk tolerance induced by personal near-loss experiences, not only from realized balance-sheet losses, regulation, or mechanical risk constraints.

2 Incremental contribution of the paper

2.1 Contribution relative to the leverage-cycle literature

- The leverage-cycle literature often explains time-varying leverage through capital constraints, intermediary balance sheets, value-at-risk rules, volatility, or collateral values.
- This paper adds a belief-based channel: lenders' personal experiences can shift collateral requirements even when balance sheets are not impaired by actual losses.
- The empirical setting is unusually clean because the lending contracts reveal haircuts directly, so leverage is measured from contract terms rather than inferred indirectly.

Interpretation: The paper's incremental step is to show that leverage can fall because lenders become more pessimistic or cautious after a near-loss episode, not because their wealth necessarily declines.

2.2 Contribution relative to experience-effects research

- Prior work shows that individuals who live through macroeconomic shocks may take less financial risk later.

- The paper extends this literature to sophisticated financial-market participants in an active collateralized lending market.
- It also shows aggregate relevance: the changed behavior of exposed lenders affects overall leverage conditions.
- The shock does not involve realized losses, so the effect is not simply an income, wealth, or balance-sheet effect.

Interpretation: The paper demonstrates that experience effects operate through perceived risk and personal salience, and can matter even among financiers in a developed market.

2.3 Contribution relative to historical finance

- The paper uses hand-collected notarial loan contracts from Amsterdam, one of the most sophisticated financial markets of the eighteenth century.
- It links contract-level lending terms to a specific financial crisis episode and to lender exposure to a failed speculative syndicate.
- It contributes a rare micro-level view of haircuts, interest rates, borrowers, lenders, collateral, and post-crisis adjustment in a premodern financial market.

Interpretation: The historical setting is not merely illustrative. It provides variation and data that are difficult to obtain in modern OTC collateral markets.

2.4 Core incremental insight

The paper's key contribution is to identify a personal-experience channel in leverage determination: lenders who nearly lost money became more conservative, while otherwise similar lenders observing the same public distress did not change behavior nearly as much.

3 Historical setting and event

3.1 Amsterdam margin lending

- Amsterdam had an active market for collateralized loans backed by securities.
- Loan contracts were negotiated over the counter, often through notaries.
- Borrowers pledged securities as collateral and received a loan for less than the collateral's market value.
- The haircut is the fraction of collateral value not financed by the loan:

$$Haircut_{i,t} = 1 - \frac{LoanValue_{i,t}}{CollateralValue_{i,t}}. \quad (1)$$

- A higher haircut means lower leverage and a larger borrower down payment.

Interpretation: The market is ideal for studying leverage because the contract directly states the amount lent and the collateral pledged, and market prices of collateral were observable.

3.2 The Seppenwolde syndicate

- The Seppenwolde syndicate was a group of Dutch investors that speculated heavily in EIC stock.
- The syndicate financed its position through collateralized borrowing.

- When EIC prices fell in 1772, lenders issued margin calls and liquidated collateral.
- The syndicate went bankrupt, but lenders recovered enough at liquidation that they did not ultimately lose money.

Interpretation: The episode creates a clean distinction between *near-loss experience* and actual wealth loss. Lenders personally exposed to the syndicate faced loss risk, but they were eventually repaid.

3.3 Treatment and control groups

- **Exposed lenders:** lenders who had extended credit to the Seppenwolde syndicate and were forced to liquidate collateral after the event.
- **Non-exposed lenders:** other lenders in the Amsterdam collateralized lending market who observed the same public market crash but did not personally lend to the syndicate.
- The key comparison is the change in contract terms after Christmas 1772 between these two groups.

Interpretation: The identifying variation is not whether a lender learned public news. Everyone learned the public news. The variation is whether the lender personally faced potential losses.

4 Data and measurement

4.1 Loan-contract data

- The loan data are hand collected from Amsterdam notarial archives.
- The main sample covers English securities pledged as collateral.
- The paper observes loan face value, collateral face value, collateral market price, interest rate, borrower identity, lender identity, and date.
- Haircuts are computed using the most recent market price of the collateral.
- Observations are often weighted by the face value of collateral.

Interpretation: The contract-level data allow the authors to measure leverage and lending terms at the level at which decisions were actually made.

4.2 Haircuts and interest rates

- The haircut is the main outcome variable.
- Interest rates are annual rates on the secured loans.
- The theoretical model predicts that pessimistic lenders will adjust primarily through haircuts, because contracts are structured to be safe for lenders.
- The empirical design therefore studies both haircuts and interest rates.

Interpretation: Finding an effect on haircuts but not interest rates supports the collateral/leverage interpretation rather than a generic increase in credit prices.

4.3 Borrower risk measures

The paper reconstructs borrower riskiness using records from the Amsterdam Bank of Exchange. Two time-varying measures are central:

$$RiskDebt_{i,t} = \log \left(\frac{debt_{i,t}}{transactions_{i,t}} \right), \quad (2)$$

$$RiskBalance_{i,t} = \log \left(\frac{balance_{i,t}}{transactions_{i,t}} \right). \quad (3)$$

- A higher debt-to-transactions ratio indicates larger collateralized debt relative to transaction activity.
- A higher bank-balance-to-transactions ratio indicates greater liquidity and therefore lower risk.
- These variables are available for a subset of borrowers but are useful robustness controls.

Interpretation: These controls address the possibility that exposed lenders lent to riskier borrowers after the crisis, which could mechanically justify higher haircuts.

4.4 Lender and borrower types

- The paper classifies lenders by status or occupation: merchants, patricians, nobles, females, and specialists.
- Borrowers are classified similarly where possible.
- Lender and borrower fixed effects are sometimes defined at the family or firm level.

Interpretation: These classifications help control for persistent differences in lending style, wealth, risk appetite, and market participation.

5 Conceptual framework

5.1 Basic logic of the model

- The paper uses an OTC collateralized lending intuition related to Geanakoplos-style leverage models.
- Optimists want to lever up and buy risky securities.
- Pessimists lend against collateral.
- The contract is designed to be risk-free for the lender by setting a sufficiently high haircut.
- More pessimistic lenders require larger haircuts.

Interpretation: The model explains why disagreement and lender beliefs matter for leverage. If a group of lenders becomes more pessimistic, those lenders reduce leverage by raising haircuts.

5.2 Key theoretical predictions

1. A secured lending contract will be structured so the lender is repaid even in bad states.
2. More pessimistic lenders charge higher haircuts.
3. Changes in lender beliefs have a larger effect on haircuts than on interest rates.
4. If only exposed lenders become more pessimistic after the crisis, their haircuts should rise relative to non-exposed lenders.

5. If the event merely changes public information about collateral, both exposed and non-exposed lenders should respond similarly.

Interpretation: The empirical test is built around the last distinction: personal exposure should matter only if the crisis changes beliefs or risk-taking through personal experience.

6 Empirical specifications

6.1 Simple difference-in-differences

The basic comparison is:

$$\Delta\Delta Haircut = (Haircut_{Post}^{Exposed} - Haircut_{Pre}^{Exposed}) - (Haircut_{Post}^{NonExposed} - Haircut_{Pre}^{NonExposed}). \quad (4)$$

Interpretation: This is the raw treatment effect: did exposed lenders change more than non-exposed lenders after Christmas 1772?

6.2 Benchmark regression

The benchmark estimating equation is:

$$Haircut_{i,t} = \beta_1 Exposed_i + \beta_2 Exposed_i \times Post1772_t + \beta_3 NonEIC_{i,t} + \varepsilon_{i,t} + \zeta_{i,t}. \quad (5)$$

- $Exposed_i$ identifies lenders who financed the Seppenwolde syndicate.
- $Post1772_t$ equals one after the Christmas 1772 event.
- $NonEIC_{i,t}$ controls for collateral other than EIC stock.
- Some specifications add year effects, lender type dummies, borrower type dummies, lender fixed effects, and borrower fixed effects.
- The key coefficient is β_2 .

Interpretation: A positive β_2 means exposed lenders required higher haircuts after their near-loss experience, relative to non-exposed lenders.

6.3 Interest-rate regression

The same specification is estimated with interest rates as the dependent variable:

$$InterestRate_{i,t} = \beta_1 Exposed_i + \beta_2 Exposed_i \times Post1772_t + \beta_3 NonEIC_{i,t} + \varepsilon_{i,t} + \zeta_{i,t}. \quad (6)$$

Interpretation: If the model is correct, the main adjustment should be in haircuts, not interest rates.

6.4 Dynamic convergence specification

To study persistence, the paper estimates:

$$Haircut_{i,t} = \beta_1 Exposed_i + \beta_2 Exposed_i \times Post1772_t + \beta_3 TimeSinceEvent_t \quad (7)$$

$$+ \beta_4 Exposed_i \times Post1772_t \times TimeSinceEvent_t + controls + u_{i,t}. \quad (8)$$

Interpretation: The coefficient β_2 captures the immediate differential increase in haircuts, while β_4 captures how quickly exposed lenders converge back toward non-exposed lenders.

6.5 Borrower-risk controls

The paper adds borrower-risk measures:

$$Haircut_{i,t} = \beta_1 Exposed_i + \beta_2 Exposed_i \times Post1772_t + \beta_3 NonEIC_{i,t} \quad (9)$$

$$+ \gamma_1 RiskDebt_{i,t} + \gamma_2 RiskBalance_{i,t} + controls + u_{i,t}. \quad (10)$$

Interpretation: This tests whether higher haircuts simply reflect lending to riskier borrowers after the crisis.

6.6 Relationship-capital specification

To test whether the effect is driven by destroyed lending relationships, the paper studies repeat matches and a Herfindahl measure of relationship lending. A representative form is:

$$Haircut_{i,t} = \beta_1 Exposed_i + \beta_2 Exposed_i \times Post1772_t + \beta_3 Herf_i \quad (11)$$

$$+ \beta_4 Herf_i \times Post1772_t + \beta_5 NonEIC_{i,t} + controls + u_{i,t}. \quad (12)$$

Interpretation: If relationship capital were the explanation, lenders with more concentrated prior relationships should show stronger post-crisis haircut increases. The paper does not find this.

7 Identification and empirical design

7.1 Why the setting is useful

- The event is sharp and well dated: Christmas 1772.
- The crisis was public, so all lenders observed the same aggregate news.
- Only some lenders were personally exposed to potential losses from the Seppenwolde syndicate.
- Exposed lenders did not actually lose money, which helps separate personal experience from wealth effects.

Interpretation: The comparison isolates personal experience with a near loss rather than public information alone.

7.2 Main identifying assumption

The core assumption is that absent personal exposure to the Seppenwolde syndicate, exposed and non-exposed lenders would have changed their haircut policies similarly after Christmas 1772, conditional on controls.

Interpretation: Figure 1 and the pre-period tables support this by showing similar pre-event haircuts and interest rates.

7.3 Threats and responses

- **Borrower risk selection:** exposed lenders may have lent to riskier borrowers. The paper controls for borrower characteristics, borrower fixed effects, and borrower-time fixed effects.
- **Collateral composition:** lenders might shift away from EIC stock. The paper controls for non-EIC collateral and studies collateral composition.
- **Relationship-capital loss:** exposed lenders may lose established borrowers. The paper tests repeat-borrower probabilities and relationship concentration.

- **Lender characteristics:** exposed lenders might be merchants, specialists, or otherwise different. The paper uses lender-type controls and interactions.
- **Attrition:** selection into the post-event sample may bias estimates. The paper examines attrition directly and re-estimates effects across selection-propensity percentiles.

Interpretation: The robustness checks are unusually detailed because the study uses historical observational data, but the pattern is highly consistent across tests.

8 Tables and figures

8.1 Figure 1: haircuts and interest rates over time

Read:

- Panel A shows that exposed and non-exposed lenders had similar haircuts before Christmas 1772.
- After the event, exposed lenders' average haircuts rise sharply, reaching almost 30 percent within six months.
- Non-exposed lenders' haircuts remain much lower.
- Panel B shows no comparable differential rise in interest rates.

Economic message: The main treatment effect appears in leverage terms, not in interest rates. This is consistent with the model's prediction that lenders adjust collateral requirements.

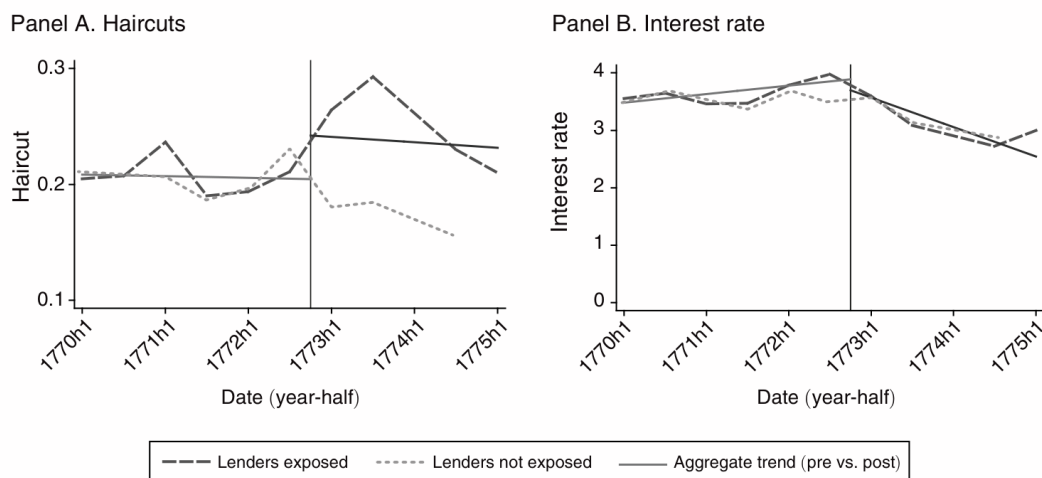


FIGURE 1. HAIRCUTS OVER TIME (*Half-Yearly Averages*)

8.2 Figure 2, Table 1, and Figure 3: the historical shock

Read:

- Figure 2 shows the EIC stock-price decline and the timing of margin calls and collateral sales around 1772–1773.
- Table 1 documents the Seppenwolde syndicate's large positions in EIC and Bank of England securities.
- Figure 3 shows that lenders liquidated collateral at positive surplus: many came close to losses, but they did not lose money.

Economic message: The event gave some lenders a personally salient near-loss experience while leaving their wealth intact. This is precisely the variation needed to test belief-based explanations.

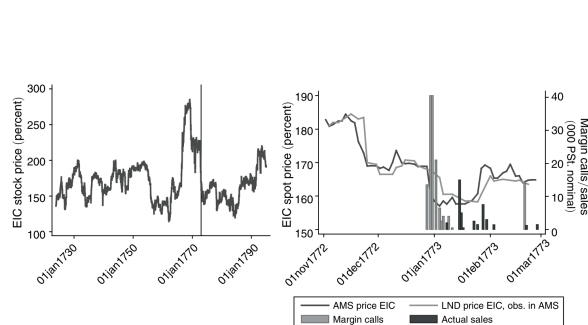


FIGURE 2. EIC STOCK PRICE IN AMSTERDAM BETWEEN 1723 AND 1793 AND EVENTS IN 1772/1773

Notes: In the left panel, the black vertical line indicates Christmas 1772. Right panel: East India Company (EIC)

Figure 2: EIC stock prices and events

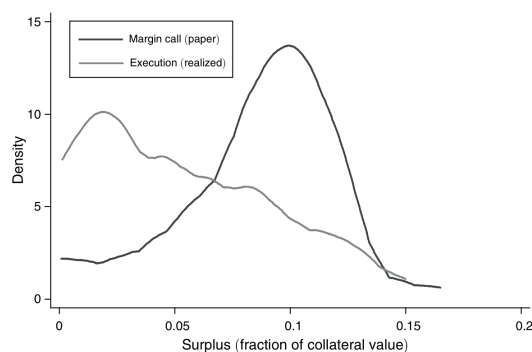


FIGURE 3. SURPLUS ON LOANS TO CONSORTIUM

Figure 3: surplus on loans to consortium

Member of the syndicate	Position (face value)	
	EIC	BoE
Hermanus van Seppenwolde	£63,600	£49,500
Johannes van Seppenwolde	£69,600	£17,000
Clifford & Chevalier	£44,500	0
Pieter van Peene	£2,000	£4,000
Total	£179,700	£70,500
Total outstanding	£3,194,080	£10,780,000
(fraction owned by syndicate)	(0.056)	(0.007)
Avg. monthly turnover (1770–1772)	£196,967	N/A
(fraction owned by syndicate)	(0.912)	N/A

Notes: Positions calculated at the end of 1772. Average monthly turnover is based on the turnover in the capital books of the respective companies. Actual market turnover would have been higher if transactions were netted out before changes in the capital books were made.

8.3 Tables 2–4: data description and pre/post sample composition

Read:

- Table 2 shows average haircuts around 20.5 percent and average interest rates around 3.63 percent.
- Table 3 shows that exposed and non-exposed lenders were broadly similar before the crisis; pre-event loan haircuts were nearly identical.
- Table 4 shows the number of loans, lenders, borrowers, and types of collateral before and after Christmas 1772.
- Exposed lenders did not simply exit the market after the crisis.

Economic message: The descriptive evidence supports the empirical design: exposed lenders were not obviously charging higher haircuts before the event, and they remained active afterward.

Variable	Observations	Mean	SD	Min	Max
Real value of collateral (guilders)	418	36,271	27,734	4,782	238,058
Face value of collateral (£)	420	1,910	1,608	300	15,000
Loan value (guilders)	422	28,969	23,244	2,200	210,000
Haircut (fraction)	418	0.205	0.059	0.080	0.550
Interest rate (percent)	420	3.63	0.30	2.50	4.00
Non-EIC (BoE, SSC, 3 percent annuities)	422	0.256	0.437	0	1

Panel A of Table 2: Descriptive information on the lenders' characteristics and data

Table 2: descriptive statistics

TABLE 4—NUMBER OF LOANS, LENDERS, AND BORROWERS, AND COLLATERAL BEFORE/AFTER CHRISTMAS 1772

	From exposed		Loans consortium			
	Yes	No	Yes	No		
Panel A. Number of loan contracts by period						
Before Christmas 1772	217	145	232	130		
After Christmas 1772	41	15		56		
	Number of lenders: exposed		Number of borrowers: from exposed			
	Yes	No	Yes	No		
Panel B. Number of lenders and borrowers by period						
Before Christmas 1772	92	82	34	38		
After Christmas 1772	18	10	21	11		
Number of new lenders/borrowers after Christmas 1772		1	12	4		
Fraction of loans from/to new lenders/borrowers		0.11	0.44	0.40		
	EIC lenders exposed		BoE lenders exposed		Other lenders exposed	
	Yes	No	Yes	No	Yes	No
Panel C. Types of collateral used by period (fractions)						
Before Christmas 1772	0.84	0.55	0.14	0.38	0.01	0.06
After Christmas 1772	0.84	0.88	0.00	0.06	0.16	0.06

Notes: Sample characteristics before and after Christmas 1772. Panel A presents the number of new loans extended by type of lender and by whether loans were taken up by the Seppenwolde consortium. Exposed lenders are those who liquidated collateral after the default. Some loans taken up by the consortium were repaid before Christmas 1772. This explains why the total number of consortium loans is larger than the number of loans extended by exposed lenders. Panel B lists the number of lenders and borrowers. We separately report the number of lenders who were exposed (yes or no). Borrowers are differentiated by whether they borrowed from exposed lenders. The panel also lists the number of new lenders and borrowers that entered the market after Christmas 1772. New borrowers are differentiated by whether they borrow from exposed lenders. We also calculate the fraction of total lending extended by new lenders/taken up by new borrowers, e.g. 44 percent of all loans extended by exposed lenders after 1772 went to new borrowers. Total lending is measured in face value of the collateral. Panel C presents the type of collateral that was used in the loan transactions, again differentiated by whether lenders were exposed (yes

TABLE 3—LENDER, BORROWER, AND LOAN CHARACTERISTICS: EXPOSED VERSUS NON-EXPOSED

	Exposed		Non-exposed		<i>t</i> -Stat
	(1)	(2)	(3)	(4)	
<i>Panel A. Lenders</i>					
Merchants (fraction total)	44	(0.537)	29	(0.433)	1.258
Patricians (fraction total)	32	(0.352)	22	(0.286)	0.909
Nobles (fraction total)	26	(0.283)	26	(0.317)	-0.493
Females (fraction total)	17	(0.185)	16	(0.195)	-0.173
Specialists (fraction total)	3	(0.033)	3	(0.037)	-0.143
<i>Panel B. Borrowers</i>					
Merchants (fraction total)	33	(0.917)	16	(0.842)	0.833
Jews (fraction total)	15	(0.417)	10	(0.526)	-0.767
Specialists (fraction total)	2	(0.078)	2	(0.105)	-0.344
$\log(\text{debt}_{i,t}/\text{transactions}_{i,t})$	3.923		3.643		0.543
$\log(\text{balance}_{i,t}/\text{transactions}_{i,t})$	1.563		1.780		-0.573
<i>Panel C. Loans</i>					
Haircuts (fraction)	0.198		0.199		-0.092
Lending volume (£ 000s)	1.890		2.064		-0.552
EIC (fraction)	0.843		0.555		5.244
Interest rate (percent)	3.763		3.526		7.773

Notes: Panel A: General characteristics of lenders who were exposed to the Seppenwolde consortium versus lenders who were not. Exposed lenders are those who were forced to liquidate collateral after the events of Christmas 1772. Merchant: active in commercial activities; patrician: member of government or the judiciary; specialist: lender who also borrows. Columns 1 and 3 report total number of lenders; columns 2 and 4 fractions of the total (non-)exposed population. Reported *t*-statistics for the difference in fractions. Panel B: general characteristics of borrowers who obtained loans from exposed or non-exposed lenders (including the Seppenwolde consortium). Merchant: active in commercial activities; $\text{debt}_{i,t}$: total collateralized debt position borrower *i* at time *t*; $\text{transactions}_{i,t}$ and $\text{balance}_{i,t}$: borrower *i*'s average daily transactions and balance in the Amsterdam Bank of Exchange during the preceding year. Panel C: general characteristics of the loans extended by exposed and non-exposed. Lending volume: measured by the face value of the collateral; EIC: fraction of loans collateralized with EIC stock. Data refer to 1770–1772 only.

Table 3: exposed vs non-exposed characteristics

8.4 Tables 5–6: main treatment effects

Read:

- Table 5 gives the simple difference-in-differences for EIC stock only: the lower-right estimate is 0.072, or 7.2 percentage points.
- Table 6 is the benchmark regression table.
- In panel A, the coefficient on $Exposed \times Post1772$ is positive across specifications, ranging roughly from 0.040 to 0.076.
- In panel B, the same interaction has no comparable significant effect on interest rates.

Economic message: The core result is robust: exposed lenders raised haircuts by several percentage points, while interest rates did not move differentially.

TABLE 5—SIMPLE DIFFERENCE-IN-DIFFERENCES ESTIMATE HAIRCUTS (EIC Stock Only)

	Before Christmas 1772	After Christmas 1772	Δ
Non-exposed	0.211	0.193	−0.018
Exposed	0.207	0.261	0.054
Δ	−0.004	0.069	0.072

Notes: Average haircuts on EIC stock, by exposed and non-exposed lenders, before and after Christmas 1772. Haircuts calculated as the fraction of the collateral value not financed with a loan. Exposed lenders: those who were forced to liquidate collateral after Christmas 1772. Each observation is a new contract. Averages are weighted by the face value of collateral. The difference-in-differences estimate is in italics in the lower-right corner.

Table 5: simple difference-in-differences

TABLE 6—BENCHMARK ESTIMATES

	OLS (1)	OLS (2)	FE (3)	FE (4)	FE (5)
<i>Panel A. Haircuts</i>					
Exposed	−0.005 (0.005)	−0.003 (0.005)		−0.000 (0.006)	
Exposed $\times$ Post 1772	0.076 (0.022)	0.066 (0.023)	0.061 (0.035)	0.040 (0.024)	0.064 (0.036)
Non-EIC	−0.059 (0.006)	−0.056 (0.006)	−0.049 (0.011)	−0.052 (0.008)	−0.047 (0.015)
Constant	0.219 (0.006)	0.245 (0.017)	0.244 (0.025)	0.211 (0.012)	0.190 (0.037)
R^2	0.334	0.440	0.630	0.659	0.798
<i>Panel B. Interest rates</i>					
Exposed	0.072 (0.036)	0.048 (0.034)		0.074 (0.041)	
Exposed $\times$ Post 1772	−0.049 (0.099)	−0.034 (0.099)	−0.080 (0.132)	0.035 (0.113)	0.066 (0.217)
Non-EIC	−0.078 (0.036)	−0.093 (0.034)	−0.085 (0.050)	−0.104 (0.049)	−0.078 (0.053)
Constant	3.527 (0.036)	3.637 (0.096)	3.674 (0.101)	3.559 (0.071)	3.792 (0.171)
R^2	0.511	0.564	0.741	0.699	0.832
Year dummies	Yes	Yes	Yes	Yes	Yes
Lender type dummies	No	Yes		Yes	
Borrower type dummies	No	Yes	Yes		
Lender FE	No	No	Yes	No	Yes
Borrower FE	No	No	No	Yes	Yes
Observations	418	387	418	387	418
Observations (if balanced)	177	152	177	152	177
Number of borrowers	72	70	72	70	72

Notes: Regression estimates for all English securities. Observations refer to new contracts and are weighted by the face value of the collateral. Haircuts are calculated as the fraction of the collateral value that is not financed with a

Table 6: benchmark estimates

8.5 Tables 7–8: borrower risk and lender risk preferences

Read:

- Table 7 checks whether exposed lenders switched to safer or riskier borrowers after the crisis.
- Borrower attrition and riskiness do not explain the main result.
- Table 8 splits or matches lenders by prior EIC exposure and interest-rate behavior.
- The treatment effect remains positive and economically meaningful in OLS and nearest-neighbor matching estimates.

Economic message: The evidence points to a change in exposed lenders’ willingness to bear risk, not merely a change in borrower pool.

TABLE 7—BORROWER ATTRITION AND RISKINESS

Panel A. Attrition of borrowers after Christmas 1772 (logit, 1 = exit)		Including consortium		Excluding consortium			
		(1)		(2)			
$\log(\text{debt}_{i,t}/\text{transactions}_{i,t})$		0.048 (0.043)		0.055 (0.061)			
$\log(\text{balance}_{i,t}/\text{transactions}_{i,t})$		−0.094 (0.056)		−0.104 (0.092)			
Observations		41		39			
Pseudo- R^2		0.063		0.043			
Panel B. Riskiness borrowers before/after Christmas 1772: exposed versus non-exposed		$\log(\text{debt}_{i,t}/\text{transactions}_{i,t})$			$\log(\text{balance}_{i,t}/\text{transactions}_{i,t})$		
		(1)	(2)	(3)	(4)	(5)	(6)
Exposed		0.452 (0.222)		0.545 (0.221)	−0.172 (0.139)	−0.311 (0.115)	
Post 1772			0.327 (0.384)	0.905 (0.448)		1.360 (0.184)	1.114 (0.514)
Exposed $\times$ Post 1772				−0.926 (0.680)		0.410 (0.532)	
Constant		3.972 (0.163)	4.193 (0.112)	3.882 (0.166)	0.930 (0.112)	0.642 (0.055)	0.819 (0.097)
Observations		342	342	342	342	342	342
Adjusted R^2		0.015	0.002	0.021	0.004	0.212	0.228

Notes: Panel A: logit estimates investigating whether a borrower drops out of the sample after Christmas 1772. $\text{debt}_{i,t}$: total margin loan position borrower i at time t . $\text{balance}_{i,t}/\text{transactions}_{i,t}$: average daily balance (transaction volume) of borrower i in the Amsterdam Bank of Exchange during the 52 weeks prior to time t . The table reports marginal effects, e.g., an increase in balance over transactions by 1 log point makes it 9.4 percent less likely that a borrower drops out of the sample. Panel B: OLS estimations at the loan level, investigating whether exposed and non-exposed lenders lent to different types of borrowers after Christmas 1772. Robust standard errors (clustered at the lender level) are reported in parentheses.

Table 7: borrower attrition and riskiness

TABLE 8—LENDERS’ RISK PREFERENCES

	EIC share			Interest		
	$<p(50)$ (1)	$\geq p(50)$ (2)	(3)	$<p(50)$ (4)	$\geq p(50)$ (5)	(6)
<i>Panel A. OLS Estimates: sample splits and interactions</i>						
Exposed	−0.003 (0.006)	−0.005 (0.008)	−0.005 (0.006)	−0.009 (0.006)	−0.009 (0.012)	−0.005 (0.005)
Exposed $\times$ Post 1772	0.048 (0.055)	0.067 (0.033)	0.056 (0.033)	0.037 (0.032)	0.093 (0.035)	0.086 (0.027)
χ^2 stat. (p -value)		0.02 (0.886)			1.52 (0.217)	
Non-EIC	−0.052 (0.009)	−0.014 (0.006)	−0.055 (0.010)	−0.060 (0.008)	−0.053 (0.012)	−0.057 (0.006)
Non-EIC $\times$ Post 1772	0.040 (0.048)	−0.021 (0.025)	0.019 (0.021)	0.023 (0.030)	0.024 (0.025)	0.021 (0.018)
EIC share/interest			0.005 (0.013)			0.009 (0.015)
EIC share/interest $\times$ Post 1772			0.041 (0.059)			−0.134 (0.061)
Objectives	188	199	384	189	198	381
Adjusted R^2	0.383	0.259	0.417	0.481	0.349	0.428
<i>Panel B. Nearest neighbor matching estimator</i>						
ATT	0.068 (0.022)	0.055 (0.022)	0.078 (0.020)	0.072 (0.026)	0.062 (0.014)	0.063 (0.012)
Additional matching variables						
Continuous	x	EIC share	x	Interest	x	x
Exact	x	Decile	Decile EIC share	Decile Interest	Decile	Deciles both
Observations	418	415	415	412	412	412

Notes: Dependent variable: haircuts. Panel A: regression estimates for all English securities, weighted by the face value of the collateral. All specifications include year fixed effects and lender and borrower type dummies. The interaction between Non-EIC and the Post 1772 dummy captures any changes in behavior in collateral ratios that

Table 8: lenders’ risk preferences

8.6 Tables 9–10: time-varying borrower risk and borrower-time comparisons

Read:

- Table 9 adds time-varying borrower-risk controls from Bank of Exchange data.
- The coefficient on $Exposed \times Post1772$ remains positive and significant in key specifications.
- Table 10 shows borrower-time fixed effect evidence for borrowers who borrowed from both exposed and non-exposed lenders after the event.
- The same borrower, at the same time, tends to receive higher haircuts from exposed lenders.

Economic message: These are especially strong checks: higher haircuts are not explained by exposed lenders facing different borrowers at different times.

	(1)	(2)	(3)	(4)	(5)
<i>Exposed</i>	0.001 (0.006)	0.002 (0.006)			−0.000 (0.006)
<i>Exposed</i> $\times$ <i>Post 1772</i>	0.052 (0.021)	0.045 (0.019)	0.101 (0.027)	0.077 (0.023)	0.054 (0.024)
<i>Non-EIC</i>	−0.050 (0.008)	−0.051 (0.008)	−0.045 (0.014)	−0.047 (0.015)	−0.053 (0.008)
$\log(debt_{i,t}/transactions_{i,t})$	0.010 (0.004)	0.010 (0.004)	0.014 (0.005)	0.015 (0.005)	
$\log(debt_{i,t}/transactions_{i,t}) \times Post 1772$		−0.017 (0.009)		−0.023 (0.012)	
$\log(balance_{i,t}/transactions_{i,t})$	−0.023 (0.007)	−0.021 (0.011)	−0.027 (0.009)	−0.026 (0.015)	
$\log(balance_{i,t}/transactions_{i,t}) \times Post 1772$		−0.017 (0.008)		−0.018 (0.013)	
Year dummies	Yes	Yes	Yes	Yes	Yes
Lender type dummies	Yes	Yes	Yes	Yes	Yes
Borrower FE	Yes	Yes	Yes	Yes	No
Lender FE	No	No	Yes	Yes	No
Borrower-time FE	No	No	No	No	Yes
Observations	317	317	341	341	381
Adjusted R^2	0.586	0.591	0.634	0.642	0.588

Notes: Dependent variable: haircuts. Regression estimates for all English securities. Observations refer to new contracts and are weighted by the face value of the collateral. Exposed lenders are forced to liquidate collateral after Christmas 1772. The interaction $Exposed \times Post 1772$ captures the difference-in-differences effect. $debt_{i,t}$: total margin loan position borrower i at time t . $balance_{i,t}(transactions_{i,t})$: average daily balance (transaction volume) of borrower i in the Amsterdam Bank of Exchange during the 52 weeks prior to time t . Lender type dummies are as in Table 3. Borrower and lender FE refer to fixed effects on the family level. Robust standard errors (clustered at the

Table 9: time-varying borrower risk

Non-exposed				Exposed			
Lender	Haircut	Face value coll.	Type	Lender	Haircut	Face value coll.	Type
<i>Panel A. Borrower Nimweegen</i>							
Boreel	0.17	2,000	P.4	Aalst	0.22	1,000	M.9
Rutgers	0.18	1,000	M.8	Graafland	0.17	1,000	P.7
			M.N.7				M.S.1
Straalman	0.18	2,500		Pereira	0.21	5,000	0
				Steenis	0.25	4,000	P.F.8
Wid. avg.	0.18			Wid. avg.	0.22		Diff.
							0.04
<i>Panel B. Borrower Gutierrez</i>							
Poorten	0.15	2,000	M.9	Berewout	0.23	1,000	M.P.1
Rutgers	0.30	1,000	M.6	Boddens	0.31	1,000	M.8
Straalman	0.30	1,000	M.N.8	Bors	0.31	2,000	N.F.8
				Hagen	0.31	2,000	N/A.6
Wid. avg.	0.23			Wid. avg.	0.30		Diff.
							0.07
<i>Panel C. Borrower Tile</i>							
Winter	0.26	1,000	M.7	Boddens	0.26	1,000	M.10
						Diff.	0.00
							0.05
				Weighted difference			

Notes: This table documents the individual transactions that drive the borrower-time fixed effects results. Information is restricted to the three borrowers who, after Christmas 1772, borrowed from both exposed and non-exposed lenders (Jan Louis van Nimweegen, Olivier Gutierrez and Penha, and Reinier Christiaan Tile). The left-hand side of the table shows lending transactions with exposed lenders, the right-hand side presents transactions with non-exposed lenders. We aggregate information by borrower-lender pair. All averages are weighted by the total face value of the collateral. $debt_{i,t}$: total margin loan position borrower i at time t . $balance_{i,t}(transactions_{i,t})$: average daily balance (transaction volume) of borrower i in the Amsterdam Bank of Exchange during the 52 weeks prior to time t . Lender type dummies are as in Table 3. Borrower and lender FE refer to fixed effects on the family level. Robust standard errors (clustered at the

Table 10: borrower-time fixed effects

8.7 Tables 11–12 and Figure 4: relationship capital, lender characteristics, and attrition

Read:

- Table 11 shows that exposed lenders were not differentially less likely to match with repeat borrowers.
- Table 12 shows that the main effect is robust to interacting the post-event period with lender types.
- Figure 4 shows that the treatment coefficient remains stable across samples defined by predicted probability of remaining in the data.

Economic message: The result is not driven by relationship-capital destruction, changing lender composition, or attrition bias.

	OLS (1)	OLS (2)	Logit (3)	Logit (4)	Probit (5)	Probit (6)
<i>Post 1772</i>	−0.211 (0.050)	−0.196 (0.108)	−0.211 (0.050)	−0.209 (0.110)	−0.211 (0.050)	−0.207 (0.104)
<i>Exposed</i>		0.026 (0.109)		0.020 (0.086)		0.021 (0.091)
<i>Exposed × Post 1772</i>		−0.018 (0.122)		−0.002 (0.196)		−0.006 (0.173)
Observations	224	224	224	224	224	224
Adj. R^2	0.046	0.037				

Notes: Dependent variable is a lender matched to a repeat borrower (one (s)he has lent to before) no = 0; yes = 1. Unit of observation: new loan contracts. To minimize measurement error of the repeat borrower variable, transactions after Jan 1, 1772 only. *Post 1772* is a dummy for contracts signed after Christmas 1772. *Exposed* is a dummy for lenders who were exposed to the Seppenwolde bankruptcy. We report marginal effects. Estimates should be interpreted as the change in the probability of being matched with a repeat borrower in response to a change in the dummy variables from 0 to 1. Robust standard errors (clustered at the lender level) in parentheses.

Table 11: repeat-borrower matching

Lender type	M (1)	P (2)	N (3)	F (4)	S (5)	All (6)
<i>Exposed</i>	−0.002 (0.005)	−0.003 (0.005)	−0.003 (0.004)	−0.003 (0.005)	−0.003 (0.005)	−0.003 (0.005)
<i>Exposed × Post 1772</i>	0.062 (0.021)	0.068 (0.023)	0.062 (0.023)	0.064 (0.022)	0.065 (0.023)	0.061 (0.021)
<i>Non-EIC</i>	−0.055 (0.006)	−0.056 (0.006)	−0.056 (0.006)	−0.056 (0.006)	−0.056 (0.006)	−0.056 (0.006)
<i>Lender type × Post 1772</i>	0.033 (0.019)	−0.015 (0.017)	−0.040 (0.019)	−0.026 (0.029)	0.005 (0.047)	
Constant	0.241 (0.018)	0.243 (0.017)	0.239 (0.017)	0.244 (0.017)	0.245 (0.017)	0.239 (0.018)
Year dummies	Y	Y	Y	Y	Y	Y
Lender type dummies	Y	Y	Y	Y	Y	Y
Borrower type dummies	Y	Y	Y	Y	Y	Y
Observations	387	387	387	387	387	387
Adjusted R^2	0.448	0.442	0.452	0.443	0.440	0.453
Lenders	152	152	152	152	152	152

Notes: Dependent variable: haircuts. Pooled OLS estimates for all English securities. Observations refer to new contracts and are weighted by the face value of the collateral. Haircuts are calculated as the fraction of the collateral value that is not financed with a loan. Exposed lenders are those who were forced to liquidate collateral after the events of Christmas 1772. The interaction between the *Exposed* and the *Post 1772* dummies captures the difference-in-differences effect. Lender and borrower type dummies are as in Table 3. Lender types: M(erchant): active in commerce; (P)atrician: member of (local) government or judiciary; N(oble), F(emale), S(pecialist): lenders also active as borrower. Column 6 contains interactions with all lender groups. Robust standard errors (clustered at the lender level) are reported in parentheses.

Table 12: lender characteristics

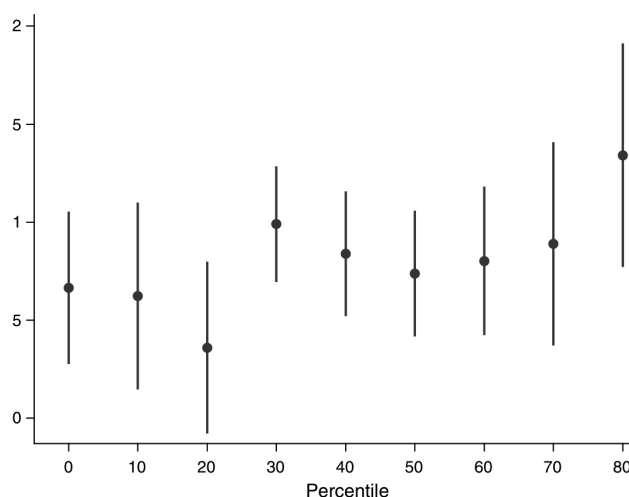


FIGURE 4 IMPACT OF ATTRITION

9 Short summary

- The paper studies eighteenth-century Amsterdam collateralized lending.
- The key event is the Christmas 1772 Seppenwolde syndicate bankruptcy.
- Some lenders were personally exposed to potential losses and forced to liquidate collateral.
- Exposed lenders ultimately did not lose money.
- After the event, exposed lenders demanded much higher haircuts.
- Non-exposed lenders, who observed the same public turmoil, changed little.
- Interest rates did not increase in parallel, so adjustment happened through collateral requirements.
- Exposed lenders also reduced lending volume and shifted toward safer borrowers.
- The effect is robust to many controls and alternative explanations.
- The paper provides micro evidence that personal experience affects beliefs, risk-taking, and aggregate leverage.

10 Conclusion

10.1 Core conclusion

Koudijs and Voth show that personal experience with a near-loss event can alter the risk-taking behavior of sophisticated financial-market participants. Lenders exposed to the Seppenwolde syndicate demanded higher haircuts after the event, even though they did not actually lose money and even though all lenders observed the same public distress.

10.2 Economic intuition

The intuition is that near-loss experience changes lenders' beliefs or subjective perception of downside risk. In a collateralized lending market, more pessimistic lenders protect themselves by increasing haircuts rather than by charging much higher interest rates. When major lenders do this, aggregate leverage falls.

10.3 Incremental contribution takeaway

The paper's distinctive contribution is to identify a personal-experience channel for leverage cycles. It separates belief changes from actual wealth losses and shows that individual risk-taking responses can aggregate into market-wide leverage changes.